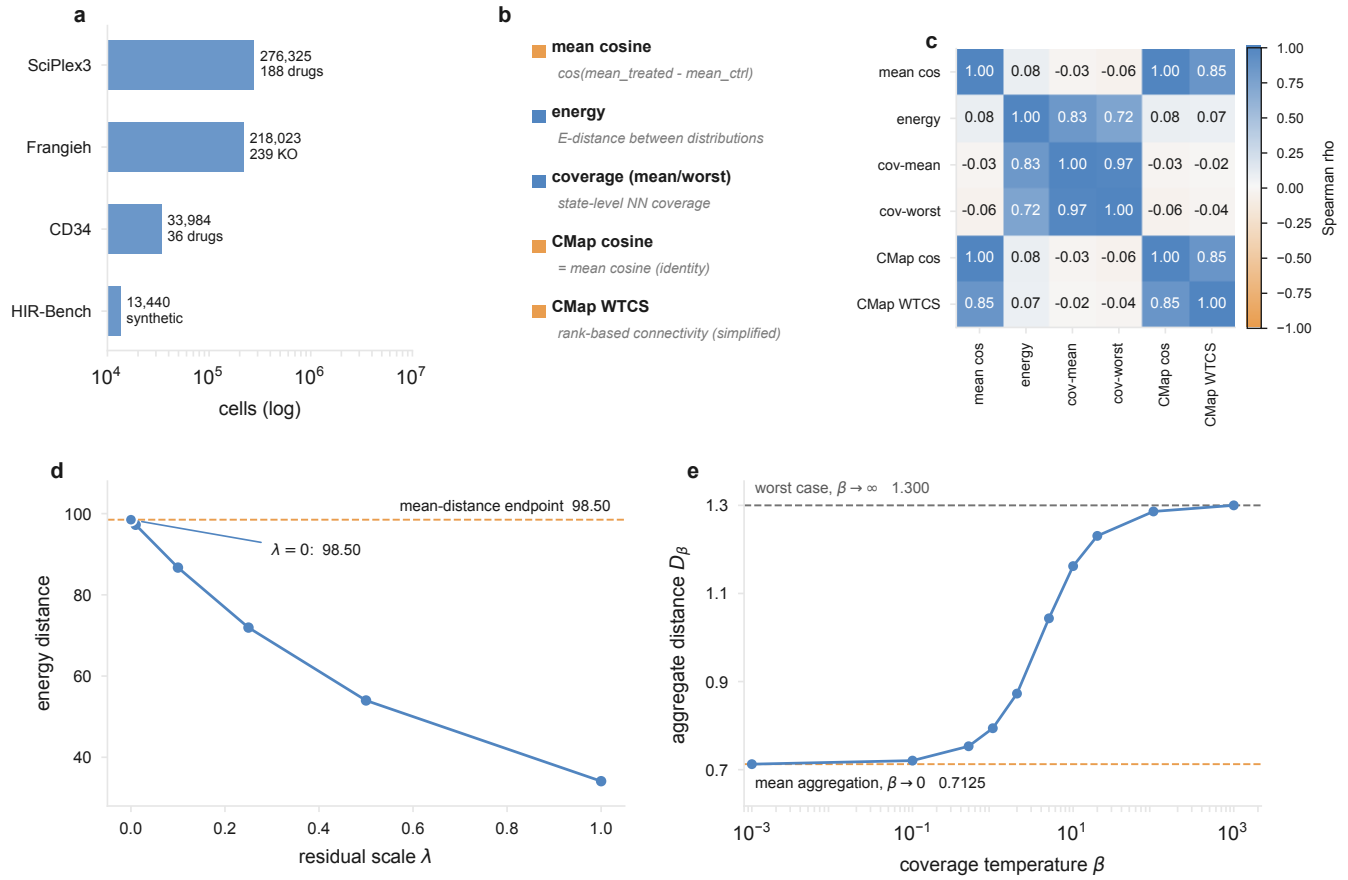
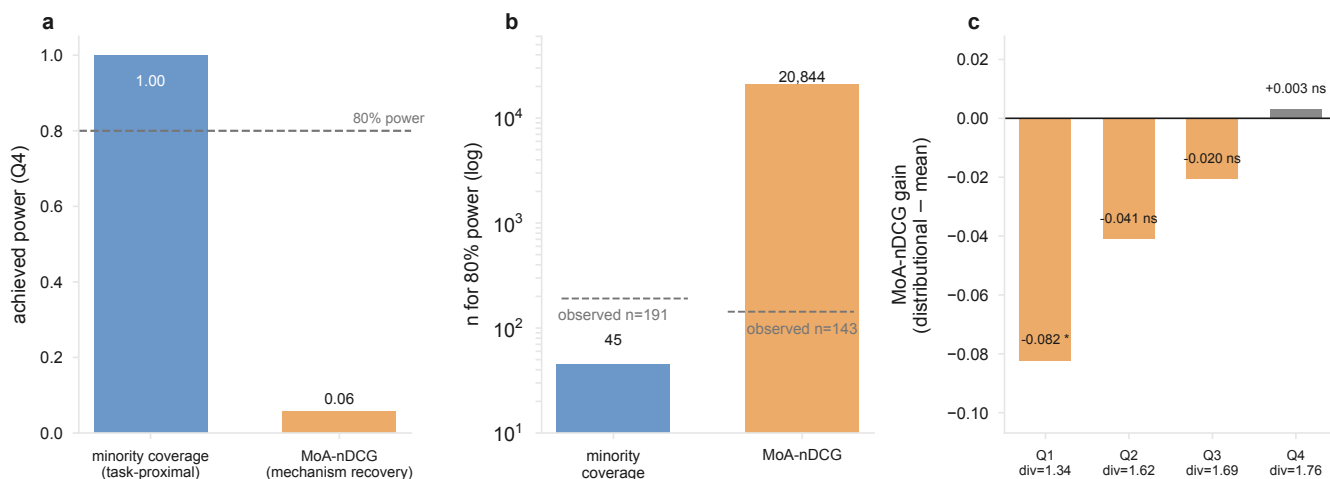


Supplementary Information for
*From signatures to populations: when does single-cell information
improve drug retrieval?*

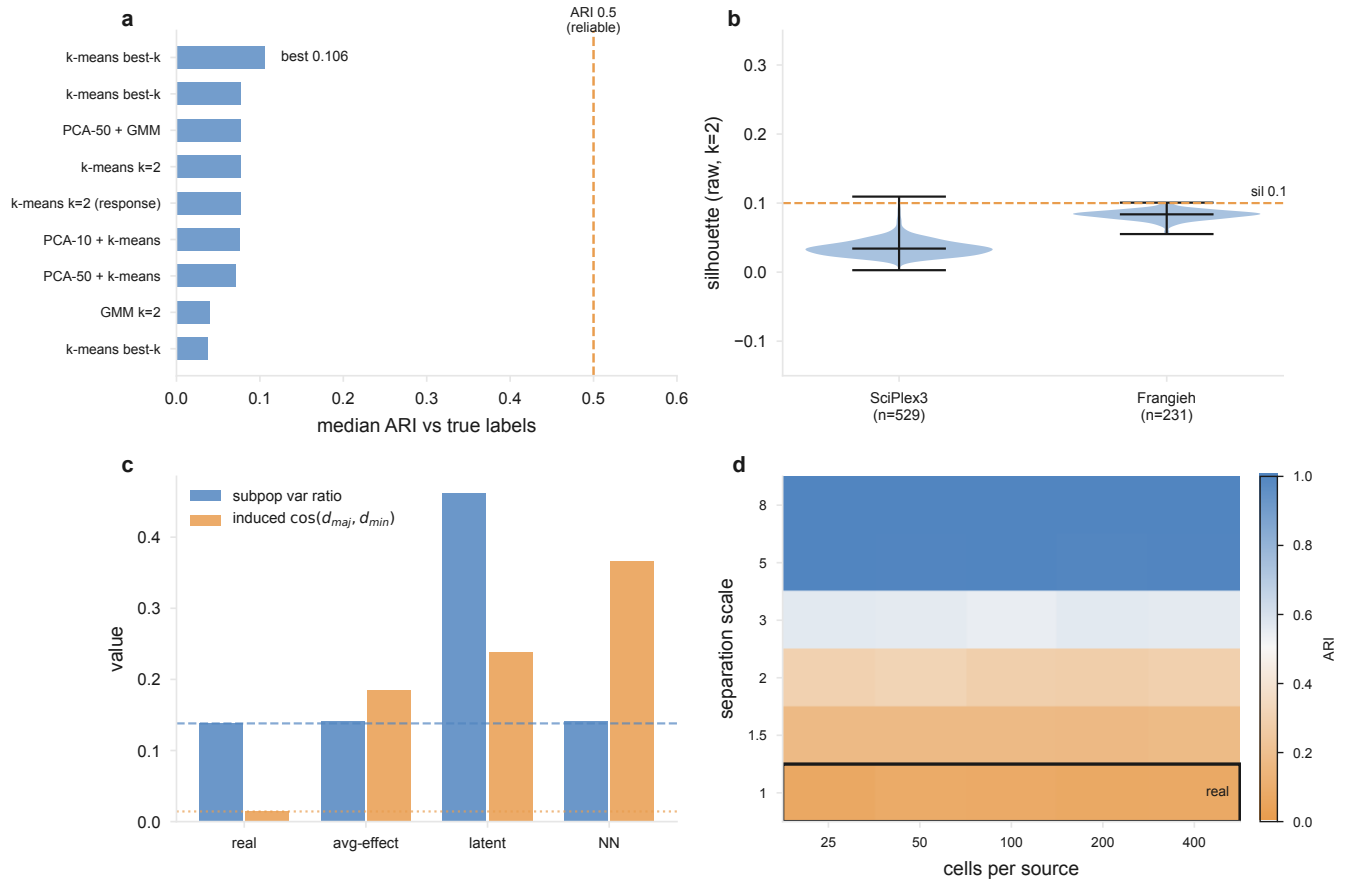
This Supplementary Information provides additional methodological detail, robustness analyses and supporting experiments for the main text. It contains seven Extended Data figures, four Supplementary Notes and four Supplementary Tables. The supplementary analyses follow the same conceptual structure as the main text: they examine the information retained by population-level retrieval, the dependence of apparent retrieval gains on the evaluation criterion, and the conditions under which state-specific single-cell responses can become consequential for candidate ranking.



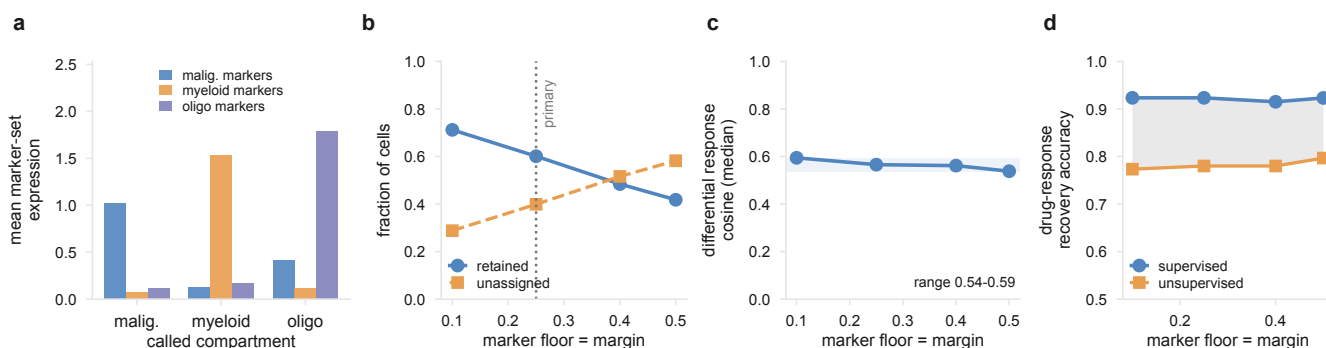
Extended Data Fig. 1. Data resources, retrieval representations, and the formal relationship between mean-signature and population retrieval. **a**, Scale of the four resources used directly in the primary retrieval and controlled synthetic analyses: SciPlex3, the primary drug-retrieval setting; Frangieh Perturb-CITE-seq, an independent genetic-perturbation setting with matched RNA-protein measurements; CD34+, a negative control; and HIR-Bench, a controlled synthetic setting. Resources used for natural heterogeneity, large-scale validation and external functional evaluation are listed in Supplementary Table 1. **b**, Retrieval representations and population-level score families. Mean-signature methods collapse each perturbation response to a control-referenced mean vector; population-level methods retain the empirical cellular response distribution. Scores evaluated: energy distance, maximum mean discrepancy, sliced Wasserstein distance and subpopulation-sensitive coverage variants, defined in Supplementary Table 2. **c**, Spearman correlation among retrieval scores over the 54,180 query-candidate scores generated by 1,260 retrieval queries. Mean cosine and the implemented CMap-style cosine agree at every one of those pairs, not merely on average ($\rho = 1.000$ to ten decimal places); the population-sensitive scores form a distinct block. These correlations describe similarity between scoring rules and are not independent statistical evidence for retrieval performance. **d**, Population-level distance converges to its mean-only endpoint as within-population variation vanishes. Writing a cellular response as $X = \mu + \lambda\epsilon$ with $E[\epsilon] = 0$, the residual of each of two synthetic populations is scaled by λ ; plotted is the energy distance between them, against the mean-to-mean distance $2\|\mu_P - \mu_Q\|$ (dashed), which is constant by construction. At $\lambda = 0$ the measured energy distance is 98.50 against a mean-distance endpoint of 98.50. Seeded synthetic construction, not a measurement on cells. **e**, One temperature interpolates mean aggregation to worst-case emphasis: the subpopulation-coverage aggregate D_β over four illustrative per-subpopulation distances, against β . Dashed rules, the two analytic endpoints, the arithmetic mean over subpopulations (0.7125, $\beta \rightarrow 0$) and the worst-matched subpopulation (1.300, $\beta \rightarrow \infty$). Energy distance is exactly the single-partition ($K = 1$) case, for which D_β equals the global energy distance at every β ; a synthetic check gives 8.229 for both.



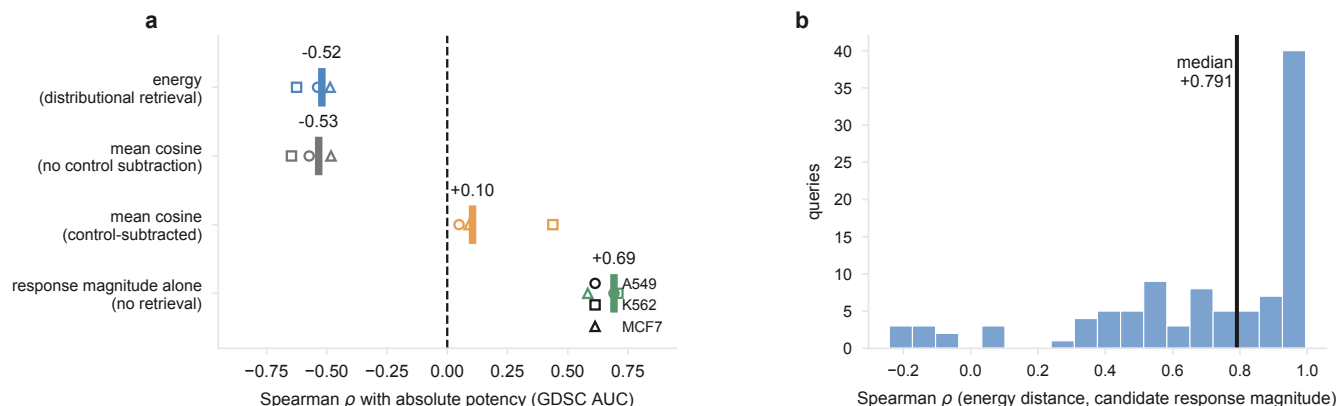
Extended Data Fig. 2. Response divergence does not reveal a hidden mechanism-recovery advantage. **a**, Achieved power for the task-proximal analyses in the highest response-divergence quartile. Minority-state coverage is well powered at the observed effect size, whereas mechanism-of-action (MoA) nDCG has low power because the observed effect is close to zero. **b**, Estimated number of queries required to reach 80% power at the corresponding observed effect sizes. **c**, Difference in MoA-nDCG between population-level and mean-signature retrieval after stratifying queries by measured response divergence. The least-divergent quartile shows a significant negative difference, whereas the median difference is zero in the remaining quartiles. The highest-divergence quartile has a small positive mean (+0.003) but no significant advantage after Benjamini–Hochberg correction ($q = 0.57$). MoA-nDCG is defined only when the query mechanism remains represented in the candidate library; 600 of the 765 partial-observation queries satisfy this requirement. Stronger state-specific response divergence therefore does not, by itself, identify a regime in which population-level retrieval improves mechanism recovery.



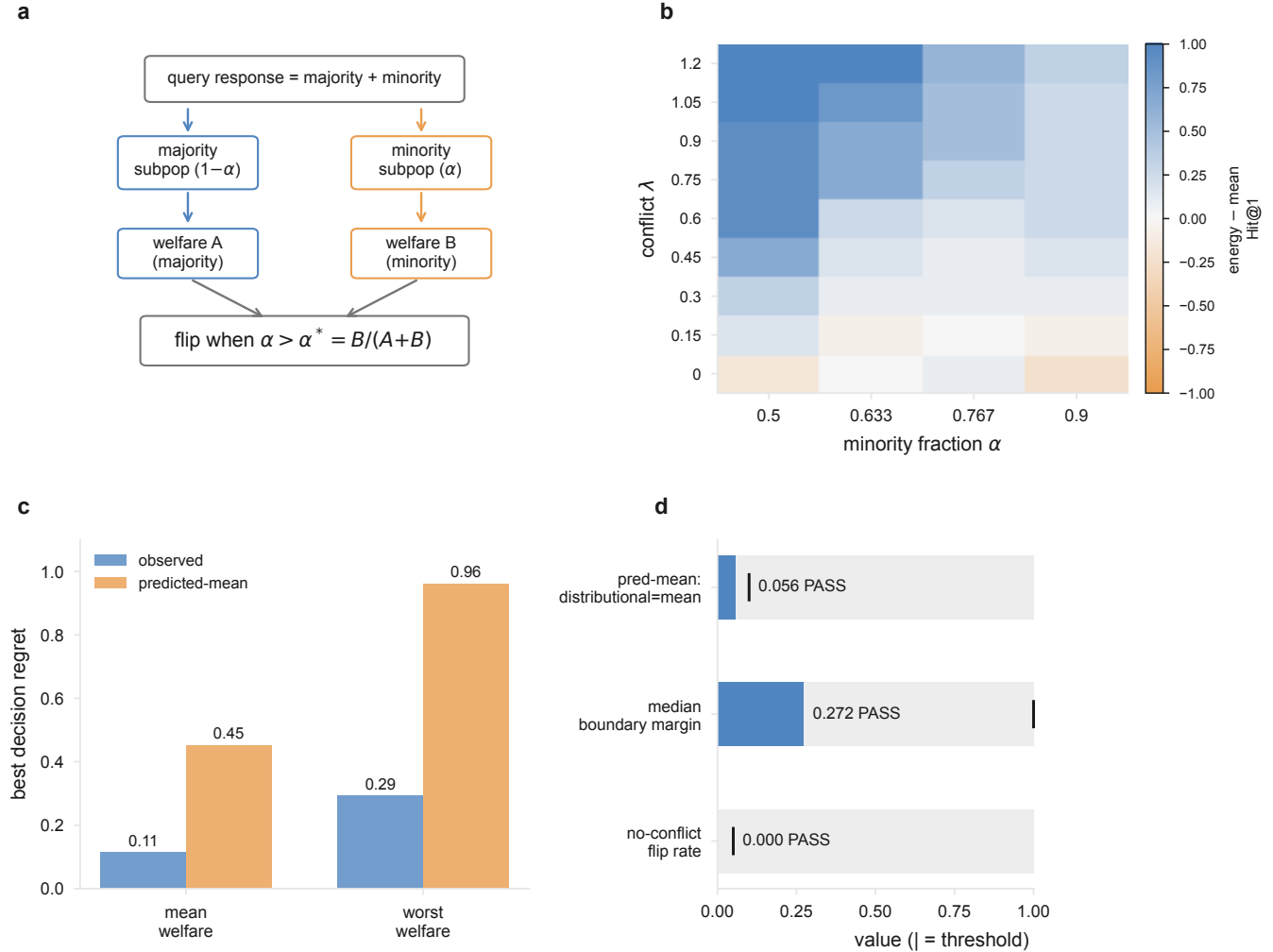
Extended Data Fig. 3. Supporting analyses of state recoverability and predicted population structure. **a**, Unsupervised recovery of the known two-source structure in the constructed K562 HDAC-JAK mixture across clustering methods and random seeds. The weak separation obtained here motivated the supervised-versus-unsupervised recoverability analysis used in the main text. **b**, Silhouette scores computed independently for perturbation-response populations in SciPlex3 and Frangieh. Most perturbation populations separate weakly under a two-cluster partition: visible multimodality is not a prerequisite for, and should not be equated with, recoverable state-specific perturbation information. **c**, Population-structure diagnostics for observed and predicted candidate populations. Blue, the subpopulation variance ratio, is flat across predictors, so baseline structure is preserved; orange, the induced-response cosine between the two subpopulations, rises, so the response divergence is not. Predicted populations can therefore retain structure inherited from the untreated population even when the predicted perturbation effect is nearly state independent. Because this panel uses cross-context constructed populations it is a structure-preservation diagnostic only; the load-bearing measurement of state-specific perturbation response is the within-context induced-response analysis in main-text Fig. 5c. **d**, Recoverability as a function of source separation and cell number in the controlled two-source mixture. Increasing source separation improves recovery, whereas increasing the cell budget alone has little effect over the range examined. The separation transform was defined from the full source populations before subsampling and was therefore independent of the particular cells used for evaluation.



Extended Data Fig. 4. Validation and threshold robustness of cellular-compartment assignments in patient-derived glioblastoma. **a**, Marker-expression validation for the three retained compartments: malignant glioma cells, tumour-associated myeloid cells and oligodendrocytes. For each called compartment, its own canonical marker set has the highest mean expression among the evaluated compartment marker sets. **b**, Fraction of cells retained or labelled unassigned as the minimum marker-expression and marker-score-margin thresholds are varied jointly from 0.10 to 0.50. The primary analysis uses 0.25 for both thresholds and excludes 39.9% of cells as unassigned. **c**, Differential-response cosine between malignant and myeloid compartments across the same threshold sweep. Median estimates remain between 0.54 and 0.59. **d**, Supervised and unsupervised recovery of within-patient drug-response structure across the threshold sweep. The supervised ceiling remains between 0.915 and 0.923 and the best unsupervised accuracy between 0.77 and 0.80, leaving a paired recoverability gap of approximately 0.11–0.13 throughout. The natural-tissue conclusions therefore do not depend on the particular 0.25 assignment threshold used in the primary analysis.



Extended Data Fig. 5. Sensitivity of external functional evaluation to response magnitude and endpoint definition. **a**, RNA-derived retrieval rankings evaluated against absolute GDSC2 drug sensitivity rather than drug-drug functional similarity. Absolute potency answers a different question from the similarity-retrieval task: a query-independent ranking based only on candidate response magnitude shows strong association with potency, whereas similarity-based retrieval scores can be weakly or negatively associated. These analyses are therefore treated as endpoint diagnostics rather than as evidence of therapeutic utility. **b**, Association between energy distance and candidate transcriptional response magnitude. Across queries, larger-response candidates tend to lie farther from other response populations (median Spearman $\rho = +0.791$). This coupling motivates the query-dependent response-magnitude-matching control used in the main GDSC2 and protein-based evaluations.



Extended Data Fig. 6. Verification and scope of HIR-Bench. **a**, Generative structure of the controlled two-subpopulation welfare model. Query responses contain majority and minority components whose preferred candidates can conflict. **b**, Difference in Hit@1 between population-level and mean retrieval across the subpopulation-balance and preference-conflict grid. **c**, Decision regret under fully observed populations, predicted populations retaining population structure and predicted populations retaining only mean-level response information. Collapsing candidate populations to their means increases regret most strongly when the evaluation criterion depends on the poorer-served subpopulation. **d**, Internal benchmark verification. The no-conflict flip rate is zero, as required by the generator, and the observed preference changes occur near the analytical boundary. HIR-Bench is a constructive demonstration that within-population structure can change candidate preference; the boundary is specified by the generator and is not a discovered biological law. Its projection to the real-data tasks (Supplementary Note 1) does not give a reliable classifier of where that happens.



Extended Data Fig. 7. Large-scale measurement of differential response, recoverability and state-specific ordering in Tahoe-100M. Tahoe-100M plate 3 contains 50 cancer cell lines, 93 compounds at a single concentration and a matched vehicle condition, giving unconstructed within-cell-line heterogeneity at substantially larger scale than the other datasets analysed here. No retrieval algorithm was run on Tahoe-100M. **a**, Differential response between two cellular states. Across 3,630 evaluable drug–cell-line conditions the median cosine between state-specific perturbation-response directions is 0.739; fewer than 0.11% of conditions are as divergent as the constructed mixtures used in the controlled analyses. **b**, Recoverability of drug-response structure. Across 960 drug pairs and 48 cell lines the supervised ceiling is 0.853 and the best unsupervised method reaches 0.611, a median paired gap of 0.157 that is positive in all 48 evaluable cell lines. **c**, Disjoint state-ordering agreement, used as a proxy for the headroom available for state-specific information to alter candidate ordering. Median Spearman correlation between drug–drug similarity matrices computed from non-overlapping cellular states is 0.841 under the cell-cycle partition and 0.778 under the independently control-derived state partition. Lower values indicate greater scope for state-specific information to change relative drug ordering; this statistic is not a direct retrieval-performance measure. **d**, Relationship between differential-response alignment and disjoint state-ordering agreement across contexts. The two measurements are related but not equivalent, reinforcing the distinction between the existence of state-specific perturbation responses and whether those differences are large enough to alter candidate ordering. Both state partitions were specified independently of downstream performance.

11 **Supplementary Note 1. Verification and transfer limits of HIR-Bench**

12 HIR-Bench was introduced as a controlled construction in which the conditions under which within-
13 population structure can alter candidate preference are known analytically. It is therefore used to verify
14 the logic of a preference-changing regime rather than to infer a biological phase boundary from data.

15 The generator satisfies its internal consistency checks on the full parameter grid. When the two sub-
16 populations have no conflict in candidate preference, the retrieval-flip rate is 0, as required by construc-
17 tion. Across the remaining parameter settings, preference changes occur near the analytical boundary

$$\alpha^* = \frac{B}{A + B},$$

18 and the median distance from the observed flip location to this boundary is 0.27. These checks con-
19 firm that the benchmark behaves according to its specified latent-welfare model.

20 We next asked whether the synthetic regime could be recognized from quantities observable at re-
21 trieval time. Observable features included the subpopulation-variance ratio, isotropy index and re-
22 sponse diversity of the query and candidate populations. Their performance is reported in the main
23 text. Features derived directly from the latent utility construction were analysed only as a controlled
24 comparison and are not available to a retrieval system.

25 The analytical boundary was also projected to the real-data retrieval tasks as a transfer test. This
26 projection did not generalize. The resulting real-data labels were strongly imbalanced (228 versus 11),
27 for which a constant majority-class predictor achieves 95.4% accuracy. The synthetic-regime projection
28 reached 61.1%, substantially below this baseline. The projection was necessarily fitted on the reduced
29 method-performance grid available for this analysis (144 instances, corresponding to 72 parameter cells),
30 rather than on the full HIR-Bench grid.

31 We therefore do not use the HIR-Bench boundary to classify biological datasets by whether pop-
32 ulation structure changes candidate preference. Its role is narrower: it establishes constructively that
33 within-population structure can become consequential for candidate choice, while the failed real-data
34 transfer shows that the corresponding regime cannot be inferred reliably from this synthetic boundary
35 alone.

36 **Supplementary Note 2. Additional diagnostics of when population information changes** 37 **ranking**

38 The main text separates the existence of state-specific perturbation responses from their recoverability
39 and from the downstream requirement that they alter candidate ordering. We performed several ad-
40 ditional analyses to ask whether simple observables could identify queries for which population-level
41 retrieval is more useful.

42 **Query identifiability.** In the SciPlex3 partial-observation tasks, query identifiability was quantified
43 from the two-state response population independently of the retrieval outcome. Across 543 evaluable
44 query–seed combinations, identifiability was positively associated with absolute mechanism-of-action
45 nDCG (Spearman $\rho = 0.17$, $p = 6.3 \times 10^{-5}$). The relationship was not monotonic, however: mean nDCG
46 across identifiability tertiles was 0.285, 0.240 and 0.444 from lowest to highest identifiability.

47 This comparison is additionally confounded by candidate-library composition. Queries in the
48 highest-identifiability tertile contained a median of six same-mechanism candidates, compared with
49 two in each of the lower tertiles, and a larger relevant candidate pool mechanically increases the
50 attainable nDCG. Even in the highest-identifiability tertile, the median silhouette score was only 0.038,
51 indicating weakly resolved two-state structure in these particular cell-line mixtures. These observations
52 therefore do not establish identifiability as a reliable predictor of population-level retrieval gain.

Measured response divergence. We next stratified the 600 mechanism-evaluable queries by the cosine between their state-specific perturbation responses. Greater divergence was associated weakly with a larger population-minus-mean mechanism-recovery difference (Spearman $\rho = 0.14$, permutation $p = 2 \times 10^{-4}$), but the conditional advantage did not become convincingly positive. The most divergent stratum reached a mean nDCG difference of approximately +0.02, with a bootstrap interval spanning zero, and the direction reversed in one of the three SciPlex3 cell lines. Stratifying by query identifiability instead yielded essentially no association with retrieval advantage.

These results distinguish differential response from decision relevance. Stronger state-specific responses may create the information required for population-aware retrieval, but they do not guarantee that this information changes the ordering of candidate perturbations.

Instability of thresholded dominance counts. We also examined whether the real-data tasks could be summarized by counting queries whose population-level gain exceeded a fixed threshold. Such counts were unstable under resampling. Using a gain threshold of +0.01, only 2 of the 10 initially threshold-crossing cross-line tasks remained above threshold in at least 8 of 10 seed resamples. Conversely, control drugs that did not cross the threshold in the recorded run crossed it in between 1 and 4 of 10 resamples.

The two most stable examples were A549→MCF7 Abexinostat (+0.090 ± 0.013 across resamples) and Belinostat (+0.081 ± 0.027), both HDAC inhibitors and both above threshold in all 10 resamples. With only two perturbations from a single drug class and one cell-line pair, these examples are treated as hypotheses rather than evidence for a general regime in which population structure changes candidate preference.

For this reason, the main text reports continuous retrieval differences and stratified analyses rather than thresholded winner counts.

Supplementary Note 3. Perturbation-predictor diagnostics

The predictor analyses in this study address a specific question rather than comparing perturbation-prediction models generally: whether a predicted response population contains state-specific perturbation effects that could be exploited by a downstream population-aware retriever.

Additive response models. For a predictor that applies the same perturbation effect δ_d to every cell in a context,

$$\hat{x}_{i,d} = x_i + \delta_d,$$

the response of any two cellular states is identical after subtraction of their corresponding controls,

$$r_{d,0} = r_{d,1} = \delta_d.$$

Consequently,

$$\cos(r_{d,0}, r_{d,1}) = 1.$$

This result is independent of the heterogeneity of the starting control population. An additive predictor can therefore generate a visually heterogeneous predicted population while containing no state-specific variation in the perturbation effect itself.

The average-effect and linear-latent predictors used in the main analysis satisfy this additive limit by construction when the perturbation response is referenced to the model-consistent baseline. Their role is to establish this limiting case, not to represent the performance of modern perturbation-prediction models as a class.

90 **scGen.** A deployed scGen model was evaluated using the same within-context cellular-state partition
91 and state-specific induced-response definition as the other predictors. Across the same 30 cell-line and
92 drug combinations, its mean induced-response cosine was 0.972, close to the additive limit of 1.000 and
93 substantially more aligned than the corresponding responses observed in real treated cells (0.205). Thus,
94 in this experiment scGen preserved substantially less state dependence in the perturbation effect than
95 was present in the observed response populations.

96 The model is the published `scgen`.SCGEN implementation on `scvi-tools`, trained on the SciPlex3
97 tensor with the perturbation label as batch key and the cell line as label key, 30 latent dimensions, batch
98 size 512, at most 40 epochs with early stopping after 15 epochs without improvement, and a fixed seed.
99 Predictions use scGen’s own latent arithmetic: one latent delta per perturbation, taken as the mean latent
100 of treated cells minus the mean latent of control cells, added to the encoded control mean and decoded
101 (Methods).

102 **Non-parametric optimal-transport mapping.** We additionally evaluated a non-parametric optimal-
103 transport map as a predictor that is not constrained to apply a shared additive shift to all cells. Under
104 the within-context differential-response analysis, the optimal-transport predictions reached an induced-
105 response cosine of approximately 0.900 ± 0.107 , moving away from the additive limit but remaining
106 much more aligned than the observed responses.

107 A latent-space implementation passed the corresponding perturbation-learning diagnostic (mean-
108 delta agreement 0.352), whereas the same construction applied directly in gene space showed substan-
109 tially weaker perturbation learning (0.057). The latent-space implementation was therefore used for the
110 differential-response comparison.

111 The map is an entropic (Sinkhorn) coupling between control and treated cells with uniform
112 marginals, regularization 0.05, at most 500 iterations and a cost matrix rescaled by its maximum, solved
113 in a 30-component PCA latent fitted unsupervised on at most 20,000 cells. Each fitting context is centred
114 on its own control mean, and the query context is excluded from the fit. A barycentric projection gives
115 a per-source-cell displacement, extended to the query’s real control cells by nearest-source matching in
116 the same centred space and decoded through the PCA inverse transform (Methods).

117 Opening this state-specific response channel was not sufficient to provide a clean retrieval test. When
118 the same optimal-transport map was used to generate candidates in the cross-context predict-then-rank
119 setting, Hit@1 was approximately 0.05 under each retrieval rule, compared with 0.88 for the additive
120 average-effect candidate source. The optimal-transport predictions were therefore too inaccurate in that
121 transfer setting to test whether greater state dependence itself improves retrieval.

122 Taken together, these analyses leave a specific unresolved experiment. A decisive test requires a
123 predictor that both generalizes accurately to the target cellular context and reproduces the state-specific
124 perturbation responses present in real cells. No predictor evaluated here satisfied both requirements
125 simultaneously.

126 **Supplementary Note 4. Robustness of the natural-heterogeneity and large-scale analyses**

127 The main text uses patient-derived glioblastoma and Tahoe-100M to ask whether the information re-
128 quirements identified in the controlled experiments are also present when cellular heterogeneity is not
129 created by combining experimentally distinct source populations. These datasets are not additional
130 retrieval benchmarks. They measure differential response, recoverability and the extent to which state-
131 specific response ordering leaves headroom for population structure to alter candidate preference.

132 **Robustness of glioblastoma compartment assignment.** The ZhaoSims2021 dataset does not provide
133 resolved cell-type labels for the analysed cells, so malignant, tumour-associated myeloid and oligoden-
134 drocyte compartments were assigned from canonical marker scores. The primary analysis required both
135 an absolute marker-expression threshold of 0.25 and a z-score margin of 0.25 over the second-highest

136 compartment score. Cells failing these criteria were excluded rather than forced into a compartment,
137 resulting in 39.9% of cells being labelled unassigned.

138 The results were stable when both thresholds were varied jointly over

$$\{0.10, 0.25, 0.40, 0.50\}.$$

139 Across this range, the excluded fraction varied from 29% to 58%, while the supervised drug-response
140 ceiling remained between 0.915 and 0.923, the best unsupervised recovery between 0.77 and 0.80, and
141 the paired recoverability gap between approximately 0.11 and 0.13. The median cosine between
142 malignant- and myeloid-specific perturbation-response directions remained between 0.54 and 0.59.
143 Thus, the natural-tissue conclusions are not specific to the primary 0.25 assignment threshold.

144 **Patient-level uncertainty.** The natural-tissue recoverability analysis contains 36 within-patient drug-
145 pair splits across four patients and two compartments, with 30 of the 36 splits contributed by a single
146 patient. Drug-pair splits were therefore not treated as independent replicates. Patient-level cluster boot-
147 strap resampling was used to quantify uncertainty, and the analysis was repeated after removing the
148 dominant patient.

149 The reported recoverability gap is computed within each split before aggregation,

$$G_j = A_{\text{sup}}^{(j)} - A_{\text{unsup}}^{(j)},$$

150 rather than as the difference between the two marginal medians. The primary paired median is
151 +0.117; subtracting the marginal medians would instead give +0.147. Across 5,000 patient-cluster boot-
152 strap resamples, the natural-minus-constructed difference in recoverability gap was +0.109 with an
153 interval of [+0.052, +0.129], and no bootstrap draw was at or below zero. Because only four patient
154 clusters contribute to this calculation, the interval is interpreted as supportive rather than as a precise
155 population-level uncertainty estimate.

156 **Large-scale measurement in Tahoe-100M.** Tahoe-100M plate 3 provides a substantially larger setting
157 in which heterogeneity occurs within cell lines rather than through constructed mixtures. Across 3,630
158 evaluable drug-cell-line conditions, the median cosine between the two state-specific response directions
159 was 0.739 (interquartile range 0.600–0.836). Fewer than 0.11% of conditions were as divergent as the
160 constructed mixtures used in the controlled experiments.

161 This estimate is not driven materially by the minimum-cell filter. Conditions were excluded when
162 one of the required state-specific treated or control arms contained fewer than 50 cells, which can prefer-
163 entially remove strong cell-cycle-arresting perturbations. Across drugs, coverage was only weakly asso-
164 ciated with induced-response cosine (Spearman $\rho = 0.077$), and restricting the analysis to well-covered
165 compounds changed the median from 0.739 to 0.740.

166 Recoverability remained measurable at scale. Across 960 sampled drug pairs and 48 cell lines, the
167 supervised ceiling was 0.853 and the best unsupervised procedure reached 0.611, giving a median paired
168 gap of 0.157. The gap was positive in all 48 evaluable cell lines. Individual unsupervised methods gave
169 similar results: *k*-means 0.592, Gaussian mixtures 0.582 and Leiden 0.563.

170 **State-specific ordering and decision-relevance headroom.** We finally compared relative drug-
171 response ordering between non-overlapping cellular states. Under the cell-cycle partition, the median
172 disjoint Spearman correlation between the two drug-drug similarity structures was 0.841 across 44
173 evaluable contexts. Under the independently control-derived two-state partition, the corresponding
174 median was 0.778 across 45 contexts.

175 These high correlations indicate that relative drug ordering is often substantially preserved between
176 cellular states even when the states respond differently. The pattern is not universal: under the control-
177 derived partition, 2 of 45 contexts had a correlation below 0.5, with a minimum of 0.236. Such con-

178 texts leave greater headroom for state-specific response information to alter candidate ordering, but this
179 statistic alone does not establish that a population-level retrieval method would improve the selected
180 candidate.

181 The two state definitions agree only moderately in their ranking of contexts (Spearman $\rho = 0.466$),
182 reinforcing that apparent decision-relevance headroom depends on how cellular states are defined. Both
183 partitions were specified independently of downstream retrieval performance, and neither was selected
184 after observing which produced the larger effect.

185 Tahoe-100M therefore supports the same separation of concepts observed elsewhere in the study:
186 state-specific responses can exist, and response identity can remain recoverable, while much of the rela-
187 tive candidate ordering is still preserved across cellular states. No retrieval algorithm was run on Tahoe-
188 100M, so these analyses constrain the conditions under which population-level retrieval could add value
189 rather than measuring such an advantage directly.

Supplementary Table 1. Data resources, preprocessing and analytical roles.

Resource	Scale	Perturbation/context structure	Representation and controls	Role in this study
SciPlex3	276,325 cells; 2,000 genes	188 compounds shared across A549, K562 and MCF7	Log-transformed 2,000-gene representation; perturbation responses referenced to matched controls	Primary observed-population drug-retrieval setting; controlled within- and cross-context constructions; source of the GDSC2-linked transcriptional rankings
Frangieh Perturb-CITE-seq	218,023 RNA-profiled cells; 2,000 genes; 20 retained surface proteins	239 CRISPR knock-outs across Control, IFN γ and Co-culture conditions	RNA and protein modalities joined by cell barcode; responses referenced to condition-matched controls	Independent genetic-perturbation retrieval setting and matched RNA-protein evaluator-shape experiment; not treated as a drug-retrieval benchmark
CD34+ HSPC	33,984 cells; 2,000 genes	36 compounds	Log-transformed expression; DMSO-referenced perturbation responses	Negative-control retrieval setting for testing whether population-level information consistently improves ranking
ZhaoSims2021 glioblastoma	10 patients; 6 compounds	Acute-slice culture and surgical biopsy with co-existing malignant, myeloid and oligodendrocyte compartments	Counts normalized to 10 ⁴ per cell, log1p-transformed and reduced to 2,000 highly variable genes; state-specific responses referenced within patient and compartment	Natural-tissue analysis of differential response, recoverability and state-specific response ordering; no retrieval benchmark is constructed
Tahoe-100M plate 3	4,158,278 cells; 2,304 genes	50 cell lines $\times$ 93 compounds at one concentration, with DMSO_TF controls	Released 2,304-gene log1p representation used without refitting; state-specific responses referenced to matched state controls	Large-scale measurement of differential response, recoverability and state-ordering agreement under intrinsic within-cell-line heterogeneity; no retrieval is run
GDSC2	286 compounds across 969 cell lines in the retained release	35 compounds matched to the SciPlex3 panel; A549, K562 and MCF7 excluded from external profiles	Fitted dose-response AUC; primary functional similarity uses cell-line-centred cross-line response profiles	External functional evaluation of fixed SciPlex3 RNA-derived rankings
HIR-Bench	13,440 synthetic instances	672 parameter configurations $\times$ 20 seeds in a two-subpopulation latent-welfare model	Synthetic observed, predicted-mean and predicted-structure information conditions	Controlled demonstration that population structure can alter candidate preference; not interpreted as a biological generative model

The resources above answer distinct questions and are not pooled as independent replicates of a single benchmark. Dataset-specific cell-number thresholds, subsampling rules and statistical units are given in the corresponding Methods sections.

Supplementary Table 2. Retrieval scores, task constructions and evaluation criteria.

(a) Retrieval scores.

Score	Input representation	Ranking rule	Role
Mean cosine	Control-referenced mean perturbation signatures	Descending cosine similarity between query and candidate mean-response vectors	Primary mean-signature baseline
CMap-style cosine	Same control-referenced mean differential-expression vectors as mean cosine	Same cosine operation as the implemented mean-cosine baseline	Signature-retrieval reference; numerically identical to mean cosine in this implementation
WTCS	Mean-collapsed differential-expression response	Rank-based weighted connectivity / enrichment score	Canonical rank-based signature baseline; related to but not algebraically identical to mean cosine
Energy distance	Empirical control-referenced single-cell response populations	Candidates ranked by increasing energy distance to the query population	Primary population-level retrieval score
RBF-MMD	Empirical response populations	Candidates ranked by increasing maximum mean discrepancy under an RBF kernel	Alternative distribution-sensitive score
Sliced Wasserstein	Empirical response populations	Candidates ranked by increasing sliced Wasserstein distance using 128 random projections	Alternative distribution-sensitive score
Subpopulation coverage	Response populations decomposed into response components	Component-level distances aggregated with temperature β ; increasing β gives progressively greater weight to the least well-matched response component	Population-sensitive score used for the partial-observation response-matching analyses

The five configured population-level scoring variants are, verbatim from the experiment registry, `energy`, `mmd`, `sliced_wasserstein`, `coverage_mean` and `coverage_worst`. The four score families become five configurations because subpopulation-coverage contributes two: coverage-mean is the $\beta \rightarrow 0$ limit of the aggregate and coverage-worst the $\beta \rightarrow \infty$ limit, both with energy distance as the per-component base metric. They are compared against `mean_cosine`, `mean_12`, `cmap_match` and `cmap_reverse`; `cmap_match` is numerically identical to `mean_cosine` on control-referenced mean vectors, and `cmap_reverse` is its negation.

Supplementary Table 2 continued. (b) Retrieval task constructions.

Task	Query and candidate construction	Candidate evaluation	Statistical unit
Observed-population source identification	A measured perturbation-response population is used as query and ranked against a library containing the perturbation that generated it and task-specific distractors	Hit@1; the true generating perturbation is relevant	Retrieval query; headline values macro-averaged over seven task-setting combinations
Controlled heterogeneous queries	Two response modes are combined at mixing weight α ; controlled SciPlex3 analyses use HDAC- and JAK-associated response sources	Source identification and response-matching criteria	Query / task setting
Partial-observation retrieval	Only the response information available under the controlled partial-observation protocol is exposed to the retriever; population and mean methods rank the identical candidate library once	The resulting fixed ranking is evaluated under response matching, MoA recovery and minority-state coverage	765 total queries
Mechanism-evaluable subset	Same partial-observation rankings; restricted to queries for which the query mechanism remains represented among candidates	MoA-nDCG	600 queries; 165 excluded because MoA recovery is undefined
Real-data minority-state projection	Top-ranked population and mean candidates evaluated on 239 task instances spanning constructed cross-line, within-line, Frangieh, CD34+ and predicted-population settings	Difference in minority-state coverage	Task-level difference; not interpreted as 239 independent biological replicates
GDSC2-linked retrieval	Held-out SciPlex3 compound at 10 μ M used as query; remaining matched compounds form a 34–35 candidate library	Spearman agreement between RNA-derived ranking and external GDSC2 functional-similarity ordering	Per-query correlation summarized within SciPlex3 cell line

Supplementary Table 2 continued. (c) Evaluation criteria applied to fixed rankings.

Criterion	Definition	Interpretation
Hit@1	Indicator that the perturbation generating the query is ranked first	Source-identification performance
Response-matching regret	Energy-based welfare regret against the latent per-cell objective used to construct the controlled partial-observation task; reported as regret reduction relative to mean retrieval	Response fidelity; intentionally close to the population information used by the retriever
MoA-nDCG	Normalized discounted cumulative gain rewarding candidates that share the query perturbation’s annotated mechanism	Task-proximal biological criterion; undefined when the relevant mechanism is absent from the candidate library
Minority-state coverage	Measures whether the selected candidate better preserves the response of the less-represented cellular state	Task-proximal measure of coverage of heterogeneous response structure
GDSC2 functional similarity	Spearman correlation between cell-line-centred cross-cell-line AUC profiles of the query and candidate drugs	Externally measured functional similarity; not absolute potency
Protein mean evaluator	Cosine similarity between control-referenced mean protein-response vectors	External evaluator preserving only mean protein response
Protein population evaluator	Negative energy distance between control-referenced per-cell protein-response populations	External evaluator retaining cell-to-cell protein-response structure
Response-magnitude matching	$-\ \vec{d}_q\ - \ \vec{d}_c\ $	Query-dependent scalar control that uses no within-population information

Query states s are obtained by k -means with $k = 2$ on the query cells, the minority state being the smaller cluster. A candidate d with response population P_d receives the welfare $U(d) = \text{agg}_s(-\mathcal{E}(P_d, Q_s))$, where $\mathcal{E}$ is the energy distance and the aggregator is the minimum over states in the worst-case form and the mean in the mean form. The response-matching regret of a selected candidate $\hat{d}$ is $\text{Regret}(\hat{d}) = \max_{d \in \mathcal{L}} U(d) - U(\hat{d}) \geq 0$, taken over the observed library $\mathcal{L}$, so the reference is the best available candidate rather than an unobservable optimum. Minority-state coverage is $\frac{1}{2}(1 + \cos(\bar{P}_d, \bar{Q}_{\min}))$, the cosine between the candidate’s mean response and the mean response of the query’s minority state rescaled to $[0, 1]$. The same definitions are given in the main Methods.

Supplementary Table 3. Perturbation-predictor implementations and their role in the differential-response analysis.

Predictor	Model form	Predicted-population construction	Use in this study	Interpretation / limitation
Average-effect	Per-drug mean perturbation delta	Predicted delta applied to the real control cells of the query context	Predict-then-rank candidate source and within-context differential-response reference	Additive by construction; state-specific induced-response cosine equals 1
Nearest-neighbour	Transfers the perturbation effect from the training context with the most similar untreated control state	Transferred effect applied to real query-context control cells	Predict-then-rank candidate source	Tests simple context transfer rather than a general virtual-cell architecture
Linear-latent	Additive latent arithmetic in an encoded representation followed by linear decoding	Observed control cells encoded, shifted in latent space and decoded individually	Predict-then-rank candidate source and additive latent reference	Can preserve baseline cellular heterogeneity while remaining additive in perturbation effect
scGen	Variational-autoencoder perturbation model with latent arithmetic	Evaluated on the same within-context state partition and state-specific response definition used for other predictors	Additional differential-response diagnostic reported in the main text	Mean induced-response cosine 0.972 over 30 cell-line and drug combinations
Optimal-transport map	Non-parametric mapping that permits cell-dependent displacements	Control cells mapped individually to predicted treated states	Additional non-additive differential-response diagnostic; also tested as a candidate-response source	Within-context induced-response cosine approximately 0.900 ± 0.107 ; cross-context candidate predictions were too inaccurate for a clean retrieval test

Supplementary Table 3 continued. Implementation details required for reproducibility.

Predictor	Required final specification
Average-effect	Drug-effect estimation set, context handling, cell cap, and exact construction used in the reported predict-then-rank run
Nearest-neighbour	Control-state similarity metric, training contexts available to each query, tie handling and transferred-response construction
Linear-latent	Encoder dimensionality, fitting set, perturbation-delta estimation, decoder definition and population-synthesis procedure
scGen	Published <code>scgen.SCGEN</code> on <code>scvi-tools</code> ; trained on the SciPlex3 tensor, batch key = perturbation, label key = cell line; 30 latent dimensions; batch size 512; at most 40 epochs with early stopping after 15 epochs without improvement; fixed seed; per-perturbation latent delta added to the encoded control mean and decoded
Optimal transport	Entropic (Sinkhorn) coupling in a 30-component PCA latent; cost rescaled by its maximum; uniform marginals; regularization 0.05; at most 500 iterations; fitted per non-query context centred on that context's own control mean, with the query context excluded; barycentric projection extended to the query's control cells by nearest-source matching and decoded by the PCA inverse transform

Predictors are used as candidate-response sources or information diagnostics rather than as a benchmark of perturbation-prediction architectures. The reported predict-then-rank experiment uses predictors fitted on the full SciPlex3 matrix and is therefore interpreted as an in-sample ceiling, not as a cross-context generalization benchmark.

Supplementary Table 4. Sensitivity of the GDSC2 functional-similarity evaluation to response-profile construction.

The primary external functional criterion compares drug–drug response profiles after removing the shared cell-line sensitivity effect. For cell line l and drug d ,

$$\widetilde{\text{AUC}}_{d,l} = \text{AUC}_{d,l} - \frac{1}{|\mathcal{D}_l|} \sum_{d' \in \mathcal{D}_l} \text{AUC}_{d',l},$$

where $\mathcal{D}_l$ contains all GDSC2 compounds measured in that cell line. Drug–drug functional similarity is then the Spearman correlation between centred cross-cell-line response profiles. The table compares this primary construction with the corresponding uncentred AUC-profile similarity.

RNA-derived ranking rule	Cell-line-centred functional similarity	Uncentred functional similarity
Energy distance	+0.276	+0.0967
Mean cosine, control referenced	+0.083	+0.138
Mean cosine, treated mean only	+0.241	+0.092
Response-magnitude matching	+0.232	+0.217
Candidate magnitude only	−0.330	+0.135
Potency matching diagnostic	+0.399	+0.290
Energy, magnitude-adjusted partial rank correlation	+0.097	n/a

Values are median per-query Spearman correlations over 103 leave-one-drug-out queries across the three SciPlex3 cell lines. The primary centred functional criterion is constructed from GDSC2 AUC profiles after subtracting each cell line’s mean response across all 286 compounds in the retained release. Drug pairs sharing fewer than 100 cell lines are excluded; 591 usable drug pairs remain, with a median of 909 shared cell lines per pair.

The change between centred and uncentred constructions shows that external evaluation is itself sensitive to how the functional measurement is summarized. Under the centred criterion, energy retrieval exceeds the control-referenced mean-signature baseline, but much of this association is also reproduced by response-magnitude matching. Under the uncentred criterion, the ordering between energy and the mean-signature baseline reverses. Neither construction is interpreted as a direct measure of therapeutic utility.